Marmara Üniversitesi
Electrical and Electronics Engineering Department
EE4091: Photonics (Spring 2023)

Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Office Hours MW: 11:30-12:30 T: 13:00-14:00	Phone ---	Email mohammad.alsunaidi@marmara	
Course Information	Text: Saleh/Teich, *Fundamentals of Photonics*	Grading	Class Work Assignments (3x5)% + Project 10%		Midterm Exam 35%	Final Exam 40%

Week	Events	Subjects	Mondays	Tuesdays
1 Feb 27-Mar 3	Course drop/add Week	The Theory of Light	Course introduction: The nature of light. Theory of light. Ray optics.	Ray optics: Prisms and Lenses
2 Mar 6-10	---		Ray optics: Lenses and Mirrors	Wave optics: Interference
3 Mar 13-17	---		Wave optics: Interference, Diffraction.	Wave optics: Diffraction.
4 Mar 20-24	Assignment 1 Due Mar 24		EM Wave optics: Maxwell's and Wave Equations. Types of waves.	EM Wave optics: Propagation.
5 Mar 27-31	---		EM Wave optics: Reflection. Refraction, Anti-reflection coating.	EM Wave optics: Polarization, Scattering.
6 Apr 1 – 9	**Mid-term Exams period**			
7 Apr 10-14	---		EM Wave optics: Light-matter interaction and the complex refractive index.	Quantum Optics. Photoelectric effect. The photon.
8 Apr 17-21	Holidays: Apr 20-23		Quantum Optics. Photoelectric effect. The photon.	Quantum Optics..
9 Apr 24-28	Labor Day: May 1	Optical Devices	Semiconductor fundamentals.	The PN Junction.
10 May 1-5	Midterm Make-up (Excused, May 2-7)		Photo-Detectors: Basics.	Photo-Detectors: pin and APD.
11 May 8-12	Assignment 2 Due May 12		Photo-Detectors: Characterization.	Photovoltaic Devices. The solar cell.
12 May 15-19	---		Light Emitting Devices (LEDs). Basics	Light Emitting Devices (LEDs). LED Structures and Characterization.
13 May 22-26	Assignment 3 Due May 12		LASERS. The Fabry-Perot Cavity.	Semiconductor Lasers.
14 May 29-June 2	Students Presentations	Special Topics	Students' Presentations.	Students' Presentations.
Jun 5 - 18	**Final Exams period**		**June 19-25 (Re-sit Final Exams period)**	